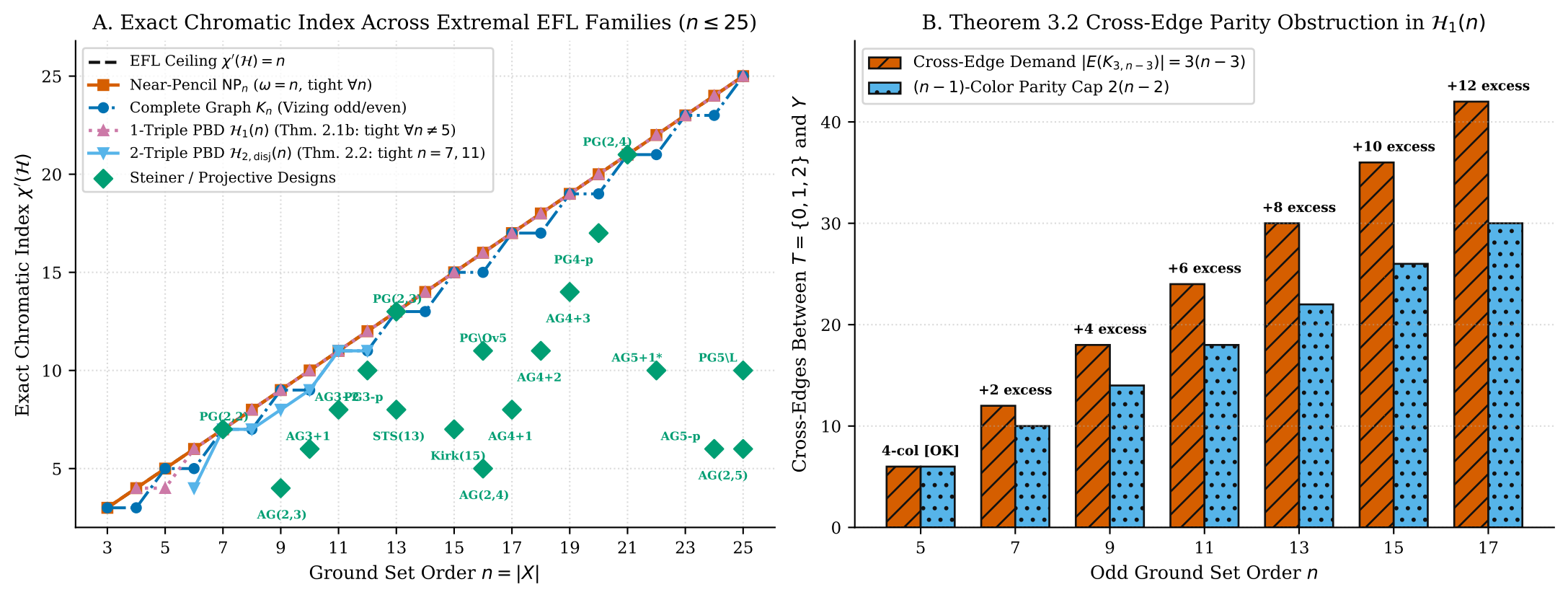

A. Exact Chromatic Index Across Extremal EFL Families ($n \leq 25$)
EFL Ceiling $\chi'(\mathcal{H}) = n$
Near-Pencil NP_n ($\omega = n$, tight $\forall n$)
Complete Graph K_n (Vizing odd/even)
1-Triple PBD $\mathcal{H}_1(n)$ (Thm. 2.1b: tight $\forall n \neq 5$)
2-Triple PBD $\mathcal{H}_{2,\mathrm{disj}}(n)$ (Thm. 2.2: tight $n = 7, 11$)
Steiner / Projective Designs
Exact Chromatic Index $\chi'(\mathcal{H})$
Ground Set Order $n = |X|$
PG(2,4)
PG4-p
AG4+3
PG(2,3)
PG\Ov5
AG4+2
AG5+1*
PG5\L
AG3+PE3-p
PG(2,2)
AG3+1
STS(13)
AG4+1
AG5-p
Kirk(15)
AG(2,5)
AG(2,4)
AG(2,3)
B. Theorem 3.2 Cross-Edge Parity Obstruction in $\mathcal{H}_1(n)$
Cross-Edge Demand $|E(K_{3,n-3})| = 3(n-3)$
$(n-1)$-Color Parity Cap $2(n-2)$
Cross-Edges Between $T = \{0,1,2\}$ and Y
Odd Ground Set Order n
4-col [OK]
+2 excess
+4 excess
+6 excess
+8 excess
+10 excess
+12 excess